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Improving coordinated actuated control performance by measuring "early return to green" using probe vehicle data

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ABSTRACT

Coordinated actuated control is designed to respond to fluctuations in traffic demand on non-coordinated phases, allowing unused green time to be reallocated to the coordinated phase. As a result, when non-coordinated phases gap out early, the coordinated green starts earlier, which is known as "Early Return to Green" (ERG). This stochastic variability across cycles adds complexity to the modeling and design of coordination strategies. Traditionally, optimizing corridors with coordinated actuated control has relied on either hard-to-access operational data from signal controllers or sophisticated optimization models that incorporate stochastic signal states. Both approaches pose significant challenges for real-world implementation. In this paper, we introduce a concise and scalable framework for incorporating ERG into the corridor modeling process when designing coordinated actuated plans. Our approach leverages probe vehicle data to estimate the distribution of actual coordination phase green start times. This distribution is then used as weights to construct a weighted probabilistic time-space diagram, which can be used to predict traffic performance in response to signal parameters. In addition, we introduce several metrics to characterize ERG, which can serve as indicators of detector failures. Numerical experiments using the SUMO simulator further validate the proposed method's accuracy against baselines and demonstrate its robustness at varying probe data penetration rates. A real-world case study demonstrates the practical application of our methodology through cycle length and offset optimization on a seven-intersection corridor under coordinated actuated control. A before-and-after evaluation highlights reductions of up to 27.3% in control delay and 16.3% in corridor travel time during peak hours.

1. Introduction

Arterial coordination plays a crucial role in improving traffic efficiency, reducing fuel consumption, and enhancing corridor-level mobility. Well-designed coordination plans time green phases across consecutive intersections in a way that promotes smoother vehicle progression and minimizes stops and delays. Effective coordination is particularly vital along major arterials, where high traffic volumes and closely spaced intersections make the system's performance highly sensitive to timing offsets. Despite extensive research on coordination optimization, most existing approaches have focused on fixed-time control systems. However, a substantial number of corridors in real-world networks operate under coordinated actuated control.

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Actuated signal control is designed to dynamically allocate green time based on traffic conditions detected by systems such as loop detectors, cameras, or LIDAR. However, current fully-actuated control strategies do not provide the same level of coordination as well-designed time-of-day coordination plans in fixed-time control. Coordinated actuated control combines these two strategies through more efficient green allocation of non-coordinated phases while maintaining the advantages of fixed coordination plans for the coordinated phases. Under this control strategy, usually only the non-coordinated phases are equipped with detection since the coordinated phases must be serviced at the same fixed point in each cycle. In this case, when the non-coordinated phases gap out or are skipped, "Early Return to Green" (ERG) occurs, in which the coordinated major phase starts earlier than this fixed point.

As a result, the start time of the coordinated green varies from cycle to cycle, depending on the gap-out behavior of non-coordinated phases, which can adversely affect coordination performance (NASEM, 2015). Although ERG increases the available green time for the major direction, it can also disrupt progression by altering the effective offsets, which are typically designed with respect to a fixed coordination reference rather than the actual green start time. Consequently, the intended coordination pattern cannot be fully realized in practice. Therefore, it is crucial to incorporate ERG considerations into the modeling and optimization of traffic coordination.

Incorporating ERG into signal coordination optimization is challenging because the start time of the coordinated phase is stochastic and depends on the service times of the non-coordinated phases. Various methods have been proposed to mitigate this issue. Jovanis and Gregor (1986) suggest using the ends of coordination at noncritical intersections as reference points to design offsets. Skabardonis (1996) employs the estimated average green start point for offset optimization. Abbas et al. (2001) propose a real-time offset adjustment algorithm that works simultaneously with ERG. Yin et al. (2007) collected archived signal status data from real-time operations to minimize the probability of vehicles encountering red lights while maximizing bandwidth. Hu and Liu (2013) use archived high-resolution signal data to estimate the conditional probability of green start times and model queue lengths to minimize total delay in real-world scenarios. Zhang and Lou (2015) incorporate actuation logic into their model and solve it using a GA-based heuristic algorithm. Ma et al. (2016) treat the coordination optimization with actuated control problem as a stochastic program and develop a rigorous formulation and solution algorithm. Mathew et al. (2019) utilize vehicle-to-infrastructure (V2I) communication data, which includes historical high-resolution signal data, to estimate green times during different time-of-day (ToD) periods.

These methods present various limitations that hinder their practical application and effectiveness. First, for approaches that use the end of the green time for coordination (Jovanis and Gregor, 1986), bandwidth utilization efficiency is often compromised. This occurs because coordinating the end of green fails to capture the densest vehicle departures, which typically occur at the green start, thereby wasting the most valuable portion of the green band. Second, methods that rely solely on the mean green start time (Skabardonis, 1996) misrepresent the influence of actuated control on traffic flow since approximating a stochastic distribution with a single mean value creates an inaccurate departure profile. In reality, the distribution of the green start acts as a smoothing filter on the departure flow. Ignoring this smoothing effect leads to significant errors in predicting downstream arrival patterns, ultimately resulting in suboptimal offset designs. Finally, approaches dependent on infrastructure data, such as archived high-resolution signal logs (Yin et al., 2007; Hu and Liu, 2013; Mathew et al., 2019) or comprehensive arrival rates for all phases (Zhang and Lou, 2015; Ma et al., 2016), face significant scalability and feasibility barriers. Accessing such data often requires specific hardware upgrades, proprietary vendor software, or advanced detection capabilities that are often infeasible on corridors with legacy infrastructure.

In this paper, we propose a method to improve coordination performance by incorporating "Early Return to Green" (ERG) using widely distributed but low-penetration vehicle trajectory data. We first introduce three metrics for quantifying ERG from these trajectory data. Next, we model ERG by exploiting departure information from partially observed trajectories. Specifically, we formulate a nonparametric maximum likelihood estimator for the probability mass function (PMF) of actual green start times for the coordination phase and solve it with an Expectation-Maximization (EM) algorithm. We then construct the weighted probabilistic time-space model to predict traffic performance metrics at various green start times, where the weights correspond to the estimated probabilities of the green start times, yielding the expected performance metrics. After constructing the model, we calibrate it using real-world trajectory data and subsequently optimize the cycle length and offsets based on the model-predicted performance metrics. Finally, we demonstrate the real-world applicability of the proposed method through a case study on a seven-intersection corridor, validating its effectiveness via before-and-after analysis.

More broadly, the proposed framework is also compatible with direct controller event logs: when such logs are available, they can be used to construct the green start PMF directly and then feed the same weighted PTS modeling, calibration, and optimization procedures. Thus, the contribution of this paper is not limited to the estimation step itself, but extends to a scalable framework for incorporating ERG into corridor modeling and optimization using whichever green start data source is available in practice.

The main contributions of this paper are as follows:

- We formulate a nonparametric maximum likelihood estimator for the distribution of actual green start times and solve it using a proposed Expectation-Maximization (EM) algorithm that fully utilizes partially observed trajectory data.
- We develop a weighted probabilistic time-space diagram to model coordinated movements with ERG, which can generate predicted performance metrics and enable calibration with real-world trajectory data.
- We introduce ERG-related metrics that can serve as practical indicators of infrastructure malfunctions, such as detector failures.
- We demonstrate the effectiveness of our method by implementing the optimized signal plan on a real-world seven-intersection corridor operating under coordinated actuated control.

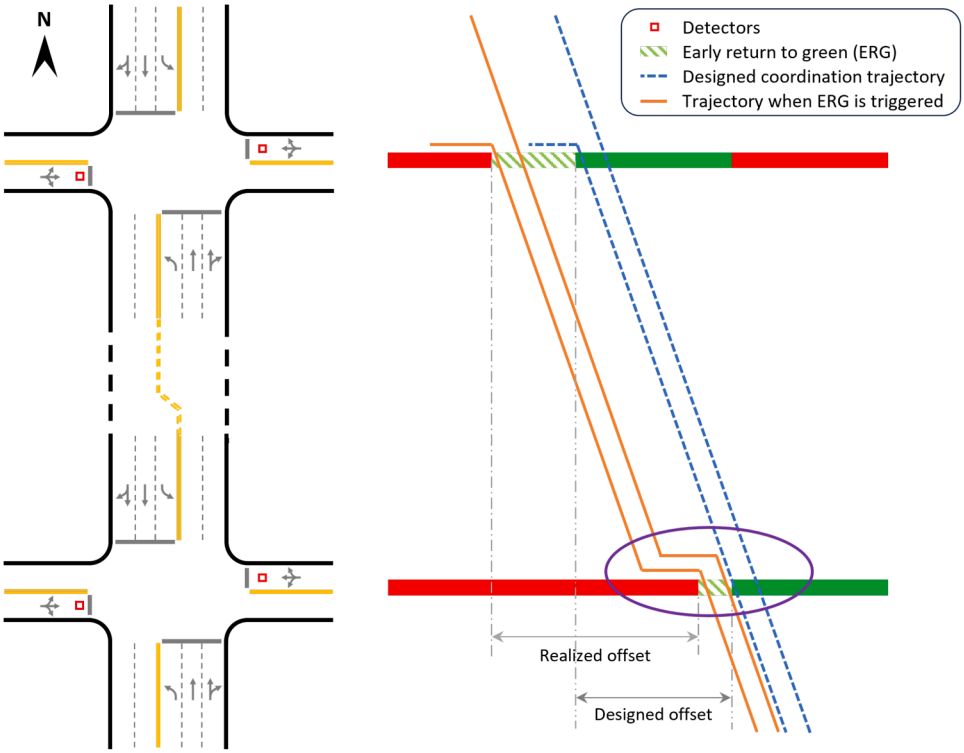


Fig. 1. Demonstration of a typical coordinated actuated road segment and its time-space diagram with early return to green (ERG).

The remainder of the paper is organized as follows. Section 2 defines the problem. Section 3 introduces the notation and the ERG-related metrics. Section 4 presents the key methodologies, including ERG modeling, the weighted PTS model and its calibration, the use of this model for cycle length and offset optimization. Section 5 details the numerical experiments conducted to validate the proposed estimation method and evaluate its sensitivity to data availability. Section 6 provides case studies, covering both individual-movement analysis and a real-world corridor implementation, and discusses the empirical results. Finally, we conclude with final remarks.

2. Problem statement

The left-hand side of Fig. 1 illustrates a typical coordinated actuated controlled segment of a corridor, where southbound and northbound represent the main directions, accounting for the majority of traffic volume at both intersections. The westbound and eastbound directions serve as the side streets. Stop bar detection can be placed on the non-coordinated phases so that the signal controller can service and extend these phases accordingly. When these phases are skipped or gap out, the remaining green time can be given to the coordinated phases. In this example, both intersections have stop bar detection on the side streets. The coordinated phases, i.e., through movements of the main direction, may commence earlier than their respective coordination start point when the side-street phases gap out or are skipped.

The side streets at both intersections may exhibit different arrival patterns, leading to different ERG patterns. However, it is currently difficult to account for these variations when designing optimal offsets, which can adversely affect coordination performance. As depicted on the right-hand side of Fig. 1, the dashed blue lines represent vehicle trajectories that follow the coordination plan, ideally proceeding downstream without stops. When ERG occurs, vehicles may encounter stops at the downstream intersection, indicating opportunities for better coordination design.

Probe vehicle data is the primary input of our method. With recent advances in connected vehicle technologies, such data has become increasingly available. For each movement, we observe a subset of all trajectories, and each trajectory is recorded at a temporal resolution of three seconds. Our previous study estimated the penetration rates to be between 6% and 10% for the study area (Wang et al., 2024). Trajectory data are well-suited for this application because of their broad coverage and simple data structure, which make them straightforward to analyze and interpret (Wang et al., 2023).

The proposed method is built on two fundamental assumptions. First, the arrival rate at non-coordinated phases is independent of the coordination offset; this simplifies the modeling of the green start time of the coordinated movement. Second, the traffic state is stationary in the analyzed ToD, and the partially observed trajectories are uniformly distributed among all trajectories for each movement; this assumption makes aggregating the trajectory data meaningful and facilitates the use of the probabilistic time-space (PTS) model. Both assumptions are reasonable and not overly restrictive, making them suitable for practical application.

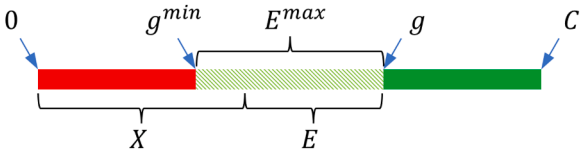


Fig. 2. Demonstration of notation.

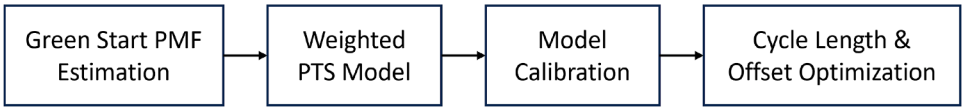


Fig. 3. Flow chart of the proposed framework for improving performance of coordinated actuated control.

3. Notation and ERG-related metrics

We introduce the following notation for use throughout the remainder of the paper. Without loss of generality, we always regard the start of the red time as the start of the cycle for the coordinated phase at a given intersection. This can be easily achieved by shifting the red start time to the left in the cycle, as illustrated in Fig. 2.

Let $g_{\min}$ and g denote the earliest possible green start time and the designated green start time for the coordinated phase, respectively. $g_{\min}$ is the summation of the guaranteed service times for all other phases in the cycle, i.e., the entire split for phases with maximum recall, the minimum green interval plus the yellow and clearance intervals for phases with minimum recall, and 0 for phases with no recall. For more information about phase modes, refer to the Signal Timing Manual (NASEM, 2015).

The actual green start time is modeled as a discrete random variable X, where $X \in \mathcal{G}$ and $\mathcal{G} = \{g_{\min}, g_{\min}+1, \ldots, g\}$ represent the set of all possible green start times, which is also the support of the probability mass function (PMF) of the actual green start time X. The ERG duration E is the difference between the actual green start time X and the designated green start time g (i.e., $E = g - X$), which is also a random variable and its realization varies among different cycles. The maximum ERG duration $E_{\max}$ is the difference between the earliest possible green start time $g_{\min}$ and the designated green start time g, i.e., $E_{\max} = g - g_{\min}$.

Based on these notations, we define several metrics that quantify the ERG pattern, using the PMF of the green start time X.

ERG occurrence rate. The ERG occurrence rate is defined as the percentage of cycles in which ERG occurs. This metric can be computed from various data sources, such as controller logs. Alternatively, it can be estimated if we have the PMF of the green start time:

$$R_{occu} = 1 - P(X = g), \tag{1}$$

where X represents the actual start time, and g is the designated green start time.

A low ERG occurrence rate might indicate a failure in the detection system or incorrect configuration settings in the traffic signal controller. Examples of such cases are discussed in the case study section.

Normalized average ERG duration and its standard deviation. We also want to measure, on average, how much ERG duration E is added to the coordination phase compared to the maximum ERG duration $E_{\max}$. Normalized average ERG duration R_{avg} is defined as the average ERG duration divided by the maximum ERG duration:

$$R_{avg} = \frac{\mathbb{E}[E]}{E_{\max}} = \frac{g - \mathbb{E}[X]}{E_{\max}}. \tag{2}$$

R_{avg} is essentially the normalized mean extra green time due to ERG. For example, $R_{avg} = 0.5$ means that the average ERG duration of the coordination phase is half of the maximum ERG duration. Moreover, following the same idea, we can define the normalized standard deviation of the ERG duration to quantify the variation of the extra green time among different cycles:

$$R_{std} = \frac{\sigma_E}{E_{\max}} = \frac{\sqrt{\mathbb{E}[(X - \mathbb{E}[X])^2]}}{E_{\max}}. \tag{3}$$

We can jointly interpret R_{avg} and R_{std} to infer side-street arrival patterns. For example, in a semi-actuated setting, a large average ERG duration (R_{avg}) combined with a small variation (R_{std}) indicates that side streets consistently experience very low arrival volumes. In contrast, when both the mean and variation are large, side-street arrivals are likely to fluctuate considerably across cycles.

These metrics can also assist traffic engineers in adjusting the side-street green split or maximum green settings. For instance, a low average ERG duration (R_{avg}) coupled with a low variation (R_{std}) suggests that the side-street phases rarely gap out and frequently max out. In such cases, increasing the maximum green time may be warranted if conditions permit.

4. Methods

To enhance coordination performance in corridors operating under coordinated actuated control, four key procedures are applied sequentially, as illustrated in Fig. 3. In this section, we describe each procedure in the order shown in the flow chart. We begin by using

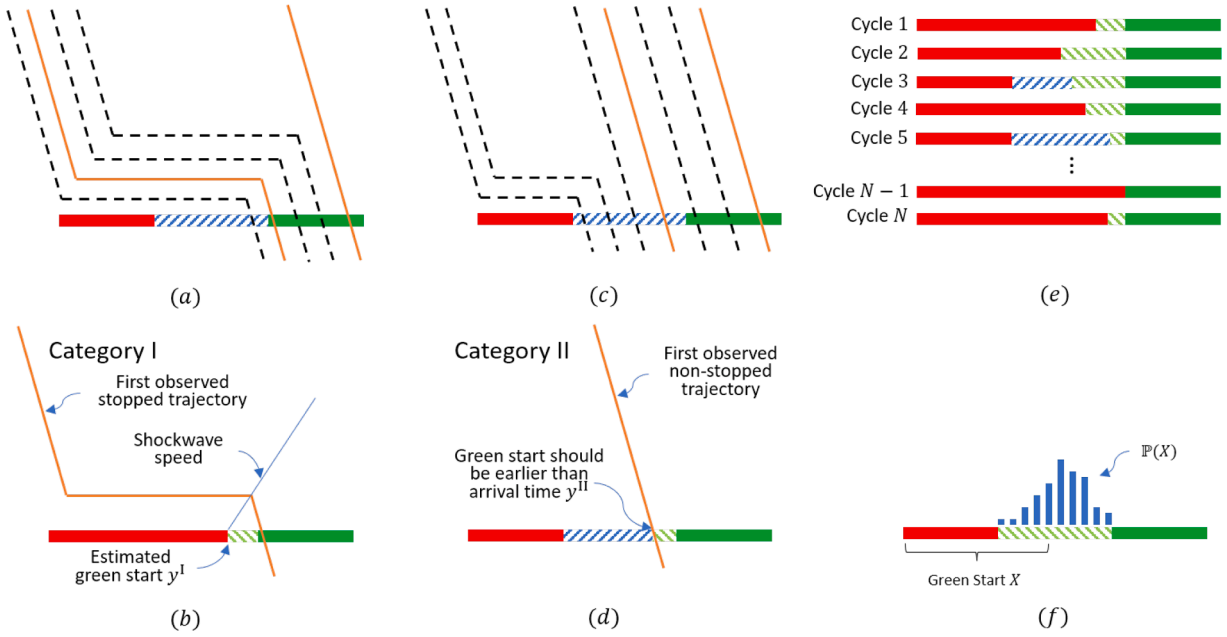


Fig. 4. Estimating the PMF of the actual green start time X.

trajectory data to estimate the distribution of the actual green start time. This distribution serves as the foundation for constructing the traffic flow model. Next, the model is calibrated using real-world trajectory data so that it can accurately predict traffic state changes under different signal plans. Finally, the calibrated model is used to optimize the signal plan for the corridor.

4.1. Estimation of actual green start time distribution

Early Return to Green (ERG) is governed by the service of non-coordinated phases operating under minimum or no recall, which depends on how frequently vehicles activate the detectors on these phases. Intuitively, modeling ERG would require considering the arrival patterns from each non-coordinated movement and then combining the results to derive the overall ERG pattern. However, trajectory data from non-coordinated movements are usually very limited since the traffic volume can be low. At the same time, we know that when a non-coordinated phase terminates early, the coordinated phase starts earlier. Therefore, under the assumption that arrivals on each non-coordinated movement are independent of the coordination offset, we adopt a simpler and more robust approach: instead of modeling the arrival patterns of all non-coordinated movements, we directly estimate the actual green start time of the coordinated phase.

Based on partially observed trajectory data, we can estimate the green start time of the coordination phase for each cycle. For this purpose, we divide the observations into two categories: cycles in which the actual green start time can be directly estimated and treated as an exact observation (Fig. 4(a)-(b)) and cycles in which only the upper bound of green start time is observed (Fig. 4(c)-(d)). Suppose both the cycle length and the designated green split are known; in this case, the solid red and green segments of the signal bar in Fig. 4 are fixed. Although the actual green start time in a specific cycle remains unknown, it must fall between the solid red and green areas, which is denoted by the blue diagonal-hatched area. In the figure, dashed black lines represent unobserved trajectories, while solid orange lines represent observed trajectories.

For the first category, where a stopped trajectory is observed, the actual green start times are treated as directly observed since we can estimate them via a simple deterministic process. Specifically, we identify the first stopped trajectory and determine its departure point, which provides both the departure time and the stopped position. From this point, we draw a line with a slope corresponding to the shockwave speed. The intersection of this line with the stop bar yields the estimated green start time y^{I} (Fig. 4(b)). As a result, the blue-hatched area in Fig. 4(a) is divided into a solid red area and a green-hatched area, representing the red phase and the additional green time due to early return to green (ERG), respectively.

The shockwave speed can be estimated by aggregating trajectories from a fixed-time intersection into a single cycle and fitting all departure points using a robust linear regression algorithm (i.e., RANSAC Fischler and Bolles, 1981). This deterministic estimation serves as the pre-processing stage of our method. Although these estimates contain minor measurement noise due to vehicle dynamics, compared to the uncertainty introduced by the second category, they are significantly more precise. Therefore, in the subsequent distribution estimation stage, we treat these estimates as exact observations for simplicity.

For the second category, only the upper bound of the actual green start times is observed since the first observed non-stopped trajectory indicates that the green phase must have started before its arrival time y^{II} at the stop bar.

By incorporating Category II observations, the blue hatched area in Fig. 4(c) shrinks from the left, resulting in an updated signal state. This means the uncertainty about the green start time is reduced compared to the case without using the non-stopped trajectory. This refinement is illustrated in Fig. 4(d).

Estimating the distribution of the actual green start time is challenging because the second category observations are left-censored: they imply $X \le y^{\mathrm{II}}$ but do not reveal X. To accommodate both exact (Category I) and censored (Category II) data within a unified framework, we formulate the problem as a Nonparametric Maximum Likelihood Estimation (NPMLE) for a discrete distribution with left-censoring (Kaplan and Meier, 1958) and develop an Expectation-Maximization (EM) algorithm to obtain the solution.

4.1.1. NPMLE formulation

Suppose we observe a total of N signal cycles. Let the index set of observations be $\mathcal{N} = \mathcal{N}_{\mathrm{I}} \cup \mathcal{N}_{\mathrm{II}}$, where $\mathcal{N}_{\mathrm{I}} \cap \mathcal{N}_{\mathrm{II}} = \varnothing$. The overall observations of the green start time are denoted as $\mathbf{y} = (\mathbf{y}^{\mathrm{I}}, \mathbf{y}^{\mathrm{II}}), \mathbf{y}^{\mathrm{I}} = \{y_i^{\mathrm{I}}\}_{i\in\mathcal{N}_{\mathrm{I}}}, \mathbf{y}^{\mathrm{II}} = \{y_j^{\mathrm{II}}\}_{j\in\mathcal{N}_{\mathrm{II}}}$. We define the feasible set S_k for the green start time in each observed cycle k as follows:

- **Category I**: For cycle i ($i \in \mathcal{N}_{\mathrm{I}}$), the first observed stopped trajectory allows us to estimate the green start time via the shockwave analysis. Let $y_i^{\mathrm{I}} \in \mathcal{G}$ denote this estimated green start time. We treat y_i^{I} as the exact observation. Thus, the feasible set is a singleton: $S_i = \{y_i^{\mathrm{I}}\}$.
- **Category II**: For cycle j ($j \in \mathcal{N}_{\mathrm{II}}$), the first observed non-stopped trajectory provides an upper bound of the green start time. Let y_j^{II} denote this upper bound on the green start time. Thus, the feasible set is defined as: $S_j = \{x \in \mathcal{G} \mid x \le y_j^{\mathrm{II}}\}$.

Let $p(x) = \mathbb{P}(X = x)$ denote the probability mass function (PMF) of X. In the nonparametric context, we are estimating the distribution $\mathbf{p} = \{p(x)\}_{x\in\mathcal{G}}$ itself. Consistent with the stationarity assumption in Section 2, we treat the green start times $\{X_k\}_{k\in\mathcal{N}}$ across different cycles as independent and identically distributed (i.i.d.) random variables. Another assumption is non-informative censoring for Category II cycles: conditional on the observed upper bound y_j^{II}, the likelihood contribution depends on X_j only through the event $X_j \le y_j^{\mathrm{II}}$. We empirically validate the robustness of our method against potential violations of this assumption using a controlled SUMO ablation experiment in Section 5 and observe accurate recovery across realistic arrival patterns and penetration rates.

Under these two assumptions, the likelihood is defined as the probability that the true green start time X_k falls within the feasible set S_k for all observed cycles. The corresponding log-likelihood is:

$$l(\mathbf{p};\mathbf{y}) = \sum_{k\in\mathcal{N}} \log \mathbb{P}_{\mathbf{p}}(X_k \in S_k). \tag{4}$$

We can now expand Eq. (4) for the two categories of feasible sets. For Category I, $\mathbb{P}(X_i \in \{y_i^{\mathrm{I}}\}) = p(y_i^{\mathrm{I}})$. For Category II, $\mathbb{P}(X_j \in S_j) = \mathbb{P}(X_j \le y_j^{\mathrm{II}}) = F(y_j^{\mathrm{II}})$, where $F(\cdot)$ is the cumulative distribution function. This leads to the final form of our log-likelihood:

$$l(\mathbf{p};\mathbf{y}) = \sum_{i\in\mathcal{N}_{\mathrm{I}}} \log p(y_i^{\mathrm{I}}) + \sum_{j\in\mathcal{N}_{\mathrm{II}}} \log F(y_j^{\mathrm{II}}). \tag{5}$$

Thus, the nonparametric maximum likelihood estimation (NPMLE) problem is formulated as

$$\begin{aligned} \max_{\mathbf{p}} \quad & l(\mathbf{p};\mathbf{y}) \\ \text{s.t.} \quad & \mathbf{p} \ge 0 \\ & \mathbf{1}^{\top}\mathbf{p} = 1. \end{aligned} \tag{6}$$

By solving this NPMLE problem, we identify the PMF that is most consistent with the constraints imposed by the two types of observations.

However, as mentioned previously, directly solving this optimization does not admit a closed-form solution because the likelihood contains censored observations (Category II). To address this, we employ the Expectation-Maximization (EM) algorithm, where the PMF $p(x)$ is iteratively updated until convergence.

4.1.2. Complete-data formulation

To apply the EM algorithm, we define the "complete data" as the precise values of the green start times for all observed cycles. Let $\mathbf{X} = \{X_k\}_{k\in\mathcal{N}}$ denote this complete set of green start times. We can partition $\mathbf{X}$ based on our observation categories:

- For Category I ($i \in \mathcal{N}_{\mathrm{I}}$), the true green start time is treated as directly observed, so $X_i = y_i^{\mathrm{I}}$.
- For Category II ($j \in \mathcal{N}_{\mathrm{II}}$), the true green start time is latent. Let Z_j denote this latent variable, where Z_j corresponds to the true green start time X_j. Although unknown, Z_j is constrained to lie within the feasible set $S_j = \{x \in \mathcal{G} \mid x \le y_j^{\mathrm{II}}\}$.

We can decompose $\mathbf{X} = (\mathbf{y}^{\mathrm{I}}, \mathbf{Z})$; thus, both y_i^{I} and z_j are realizations of X_i and X_j.

The complete-data log-likelihood is simply the joint probability of these green start times occurring:

$$l_c(\mathbf{p};\mathbf{y}^{\mathrm{I}},\mathbf{Z}) = \sum_{i\in\mathcal{N}_{\mathrm{I}}} \log p(y_i^{\mathrm{I}}) + \sum_{j\in\mathcal{N}_{\mathrm{II}}} \log p(z_j). \tag{7}$$

4.1.3. E-step

Let $p^{(t)}$ and $F^{(t)}$ denote the PMF and CDF at iteration t, respectively. Let $\mathbf{p}^{(t)}$ represent the entire distribution vector at iteration t accordingly. The Q-function is defined as the expectation of the complete-data log-likelihood with respect to the conditional distribution of the latent variables given the observed feasible sets and the current distribution $\mathbf{p}^{(t)}$:

$$Q(\mathbf{p} \mid \mathbf{p}^{(t)}) = \mathbb{E}_{\mathbf{Z}|\mathbf{y},\mathbf{p}^{(t)}}[l_c(\mathbf{p}; \mathbf{y}^{\mathrm{I}}, \mathbf{Z})]. \tag{8}$$

Since the first term in Eq. (7) contains no latent variables, it remains unchanged. For the second term, the observation y_j^{II} implies the feasible set $S_j = \{x \in \mathcal{G} \mid x \le y_j^{\mathrm{II}}\}$, so we compute the expectation of the log-probability of Z_j given the constraint $Z_j \in S_j$. Thus, the conditional probability mass function is the truncated distribution of $p^{(t)}(z)$ over the feasible range:

$$\mathbb{P}(Z_j = z \mid Z_j \in S_j) = \begin{cases} \frac{p^{(t)}(z)}{F^{(t)}(y_j^{\mathrm{II}})} & \text{if } z \le y_j^{\mathrm{II}} \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

Substituting this into the expectation yields:

$$\mathbb{E}\left[\sum_{j\in\mathcal{N}_{\mathrm{II}}} \log p(z_j)\right] = \sum_{j\in\mathcal{N}_{\mathrm{II}}} \sum_{z\in S_j} \frac{p^{(t)}(z)}{F^{(t)}(y_j^{\mathrm{II}})} \log p(z). \tag{10}$$

Therefore, combining both categories yields

$$Q(\mathbf{p} \mid \mathbf{p}^{(t)}) = \sum_{i\in\mathcal{N}_{\mathrm{I}}} \log p(y_i^{\mathrm{I}}) + \sum_{j\in\mathcal{N}_{\mathrm{II}}} \sum_{z\in S_j} \frac{p^{(t)}(z)}{F^{(t)}(y_j^{\mathrm{II}})} \log p(z). \tag{11}$$

Let $h_{\mathrm{I}}(x)$ denote the number of times we observe the exact value x in Category I data. Grouping terms by each possible $x \in \mathcal{G}$, the Q-function can be rewritten as

$$Q(\mathbf{p} \mid \mathbf{p}^{(t)}) = \sum_{x\in\mathcal{G}} \left(h_{\mathrm{I}}(x) + \sum_{j\in\mathcal{N}_{\mathrm{II}}, x\in S_j} \frac{p^{(t)}(x)}{F^{(t)}(y_j^{\mathrm{II}})} \right) \log p(x) \tag{12}$$

The term in parentheses represents the expected count of cycles corresponding to green start time x, with contributions from both Category I and Category II data. We can define this expected count of green start time x as

$$h^{(t)}(x) = h_{\mathrm{I}}(x) + \sum_{j\in\mathcal{N}_{\mathrm{II}}, x\in S_j} \frac{p^{(t)}(x)}{F^{(t)}(y_j^{\mathrm{II}})}. \tag{13}$$

4.1.4. M-step

We have the final form of our maximization problem:

$$\begin{aligned} \max_{p(x), x\in\mathcal{G}} \quad & \sum_{x\in\mathcal{G}} h^{(t)}(x) \log p(x) \\ \text{s.t.} \quad & \sum_{x\in\mathcal{G}} p(x) = 1 \end{aligned} \tag{14}$$

Eq. (14) is a classic problem of estimating probabilities from counts. By solving this with the method of Lagrange multipliers, we have

$$p^{(t+1)}(x) = \frac{h^{(t)}(x)}{\sum_{x\in\mathcal{G}} h^{(t)}(x)} \tag{15}$$

which means the PMF is then updated by normalizing the expected counts.

4.1.5. Solution algorithm and implementation notes

Although the derivation appears complex, the implementation is straightforward. In the E-step, Category II observations contribute to the expected counts via Eq. (13); in the M-step, the PMF is updated by normalization according to Eq. (15). Each step of the proposed EM algorithm is illustrated in Fig. 5. This EM procedure is repeated until the PMF converges, as shown in Fig. 4(f). The full procedure is summarized in Algorithm 1.

It is worth noting that for the initialization step, if Category I observations are sparse, utilizing a uniform distribution to initialize the PMF (i.e., $p^{(0)}(x) = 1/|\mathcal{G}|$) presents a more robust alternative. Although the current approach of employing the Category I histogram accelerates convergence, uniform initialization ensures that no feasible green start times are assigned zero probability a priori. This prevents the algorithm from prematurely excluding feasible states and converging to a suboptimal solution due to the absence of initial observations.

Note that if the control modes for some of the non-coordinated phases are *minimum recall*, i.e., the phases cannot be skipped, then the PMF obtained from Algorithm 1 can be directly used as the green start time PMF for the coordinated phase.

However, if all of the non-coordinated phases operate with *no recall*, i.e., these phases can be skipped, then the PMF must be modified accordingly. In this case, the minimum green start time is $g_{\min} = 0$, but when the no recall phases are called, there exists a

Algorithm 1 EM algorithm for estimating PMF of discrete green start time X.

Input: Category I observations $\{y_i^{\mathrm{I}}\}_{i\in\mathcal{N}_I}$, Category II observations $\{y_j^{\mathrm{II}}\}_{j\in\mathcal{N}_{\mathrm{II}}}$, $y_i^{\mathrm{I}}, y_j^{\mathrm{II}} \in \mathcal{G}$

Initialization

$h_{\mathrm{I}}(x) \leftarrow \sum_{i\in\mathcal{N}_{\mathrm{I}}} \mathbb{1}_{x=y_i^{\mathrm{I}}}$ for all $x \in \mathcal{G}$ ▷ *Initialize expected count w/ Cat. I obs.*

$p^{(0)}(x) \leftarrow h_{\mathrm{I}}(x) \big/ \sum_{x'\in\mathcal{G}} h_{\mathrm{I}}(x')$ ▷ *Initialize PMF*

$t \leftarrow 1$

repeat

  $h^{(t)}(x) \leftarrow h_{\mathrm{I}}(x)$

  for $j \in \mathcal{N}_{\mathrm{II}}$ **do**

    for $x = g_{\min}$ **to** y_j^{II} **do**

      $h^{(t)}(x) \leftarrow h^{(t)}(x) + p^{(t-1)}(x) \big/ \sum_{x'\le y_j^{\mathrm{II}}} p^{(t-1)}(x')$ ▷ ***E-step**: Update expected count*

  $p^{(t)}(x) \leftarrow h^{(t)}(x) \big/ \sum_{x'\in\mathcal{G}} h^{(t)}(x')$ ▷ ***M-step**: Update PMF*

  $t \leftarrow t + 1$

until convergence

Output: Final estimated PMF $p^{(t)}(x)$

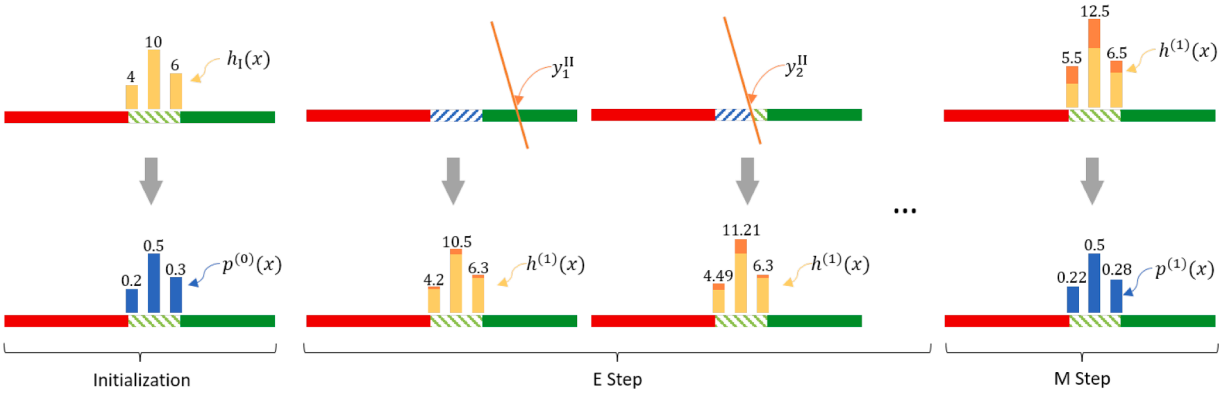


Fig. 5. Illustration of the proposed EM algorithm steps. Initialization: The process begins with counts $h_I(x)$ derived solely from stopped vehicles (Category I), yielding an initial PMF $p^{(0)}(x)$. E-Step: For each observation in Category II (e.g., $y_1^{\mathrm{II}}, y_2^{\mathrm{II}}$), the algorithm updates the expected count of the green start time x given the constraint $X \le y^{\mathrm{II}}$ by adding "fractional counts" (shown in orange) to the existing counts. Note that for observation y_2^{II}, counts are only added for $x \le y_2^{\mathrm{II}}$ (i.e., first and second bins), reflecting the censored nature of the data. M-Step: The total expected counts $h^{(1)}(x)$ are normalized to produce the updated PMF $p^{(1)}(x)$.

minimum green time for those phases. We denoted the sum of the minimum green time and the clearance interval as $g'_{\min}$. For the coordinated phase, the updated set of possible green start times is defined as $\mathcal{G}' = \{0, g'_{\min}, g'_{\min}+1, \dots, g\}$. The PMF computed from Algorithm 1 is defined on $\mathcal{G} = \{g_{\min}, g_{\min}+1, \dots, g\} = \{0, 1, \dots, g\}$, which includes values between 0 and $g'_{\min} - 1$ that are not valid under the new support $\mathcal{G}'$.

To resolve this support mismatch, we reassign the probabilities associated with $x \in [0, g'_{\min} - 1]$ to $x = 0$, which represents the case where the non-coordinated phase is skipped. Let $p'(x)$ denote the adjusted PMF defined on $\mathcal{G}'$; we have:

$$p'(x) = \begin{cases} \sum_{x'=0}^{g'_{min}-1} p(x'), & \text{if } x = 0 \\ p(x), & \forall x \in \mathcal{G}' \setminus \{0\} \end{cases}. \tag{16}$$

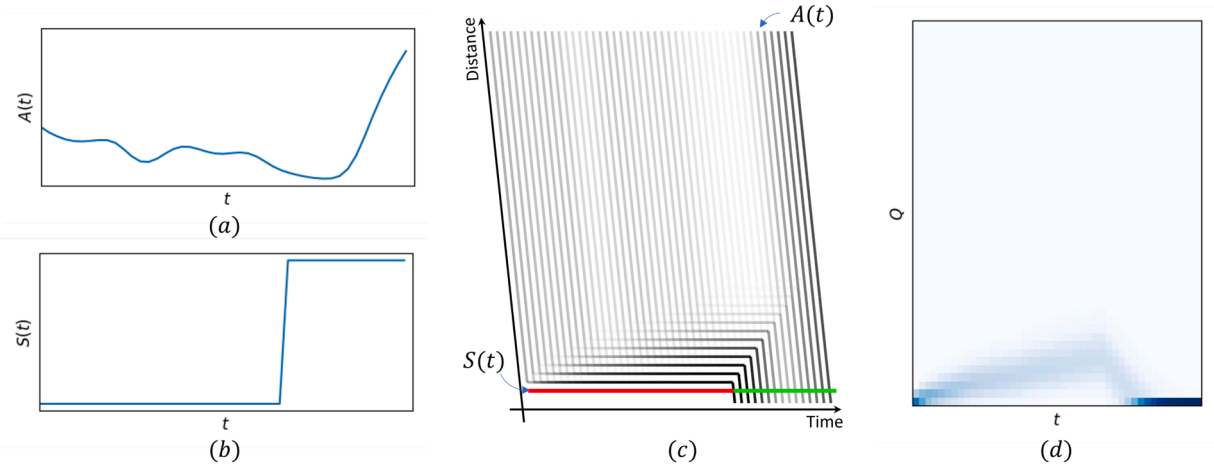


Fig. 6. Demonstration of the PTS model. (a) Arrival rate profile $A(t)$; (b) Signal state $S(t)$; (c) Probabilistic time-space diagram given the arrival rate and signal state, plotted in Newellian coordinates, with probabilities mapped linearly to opacity; (d) Heatmap of the point queue probability mass function (PMF) $P(Q,t)$.

4.2. Weighted probabilistic time-space diagram

The probabilistic time-space (PTS) model[1] is a recently developed stochastic traffic flow model that describes the traffic flow dynamics at signalized intersections with a queueing model (Wang et al., 2024). In this model, each vehicle is assumed to have only two possible states: traveling at a predefined free-flow speed v_f or stopping in the queue waiting for the traffic signal. Intermediate states and heterogeneous driving behaviors are ignored. As shown in Fig. 6(c), the PTS model can be visualized as a PTS diagram, which resembles a conventional time-space (TS) diagram. However, unlike the TS diagram where each line represents a trajectory, the opacity of a line in the PTS diagram represents the probability of a vehicle being present.

The PTS model accepts two primary inputs: the arrival rates and the traffic signal states within a cycle, as shown in Fig. 6(a)-(b). The cycle is discretized into time steps $t = 1, \dots, C$, where C is the cycle length. The arrivals are denoted $\alpha = (\alpha_1, \dots, \alpha_C)$, and the signal states are denoted $\delta = (\delta_1, \dots, \delta_C)$. Here, $\alpha_t \in [0,1]$ represents the Bernoulli arrival rate at time t, and $\delta_t \in \{0,1\}$ indicates the signal state at time t (0 = red light, 1 = green light). The PTS model outputs the point queue probability mass function (PMF) at each time step t (Fig. 6(d)). Let Q_t be the random variable representing the point queue length at time t. The output point queue PMF is $(\pi_t)_{t=1}^{C} = \mathrm{PTS}(\alpha, \delta)$, where $\pi_t(q) = \mathbb{P}(Q_t = q)$ for $q \in \{0, \dots, Q_{\max}\}$, and $Q_{\max}$ is the maximum queue length that can be observed.

This point queue PMF enables the computation of performance metrics such as stop delay and number of stops.

Average number of stops. Let $n(\alpha, \delta)$ denote the average number of stops given the arrival α and signal plan δ. We have

$$n(\alpha,\delta) = \frac{\sum_{t=1}^{C} \alpha_t[(1-\delta_t) + \delta_t(1-\pi_t(0))]}{\sum_{t=1}^{C} \alpha_t}, \tag{17}$$

where the numerator counts stops in two cases: (1) during red ($\delta_t = 0$), all arrivals stop; (2) during green ($\delta_t = 1$), arrivals stop only if the queue is non-empty, which occurs with probability $1 - \pi_t(0)$. The denominator is the total number of arrivals in the cycle.

Average stop delay. The average stop delay $d(\alpha, \delta)$ is computed as:

$$d(\alpha,\delta) = \frac{\sum_{t=1}^{C} \sum_{q=1}^{Q_{\max}} \sum_{q'=q}^{Q_{\max}} \pi_t(q')}{\sum_{t=1}^{C} \alpha_t}, \tag{18}$$

where the numerator sums, over all time steps and queue positions, the probabilities that a stopped vehicle occupies that position.

Adaptation to ERG scenarios. The original PTS model was developed for fixed-time signal control, where the signal state is deterministic. With a simple modification, it can also model movements with the potential for early return to green (ERG) at coordinated actuated intersections. Let δ^x denote the signal state sequence corresponding to an actual green start time x, and let X be the discrete random variable representing the green start time. The model-estimated average stop delay is

$$d^{\mathrm{est}} = \mathbb{E}_X\left[d(\alpha, \delta^X)\right] = \sum_{x \in \mathcal{G}} p(x) d(\alpha, \delta^x), \tag{19}$$

[1] For completeness, a summary of the mathematical formulation of the PTS model is provided in Appendix A

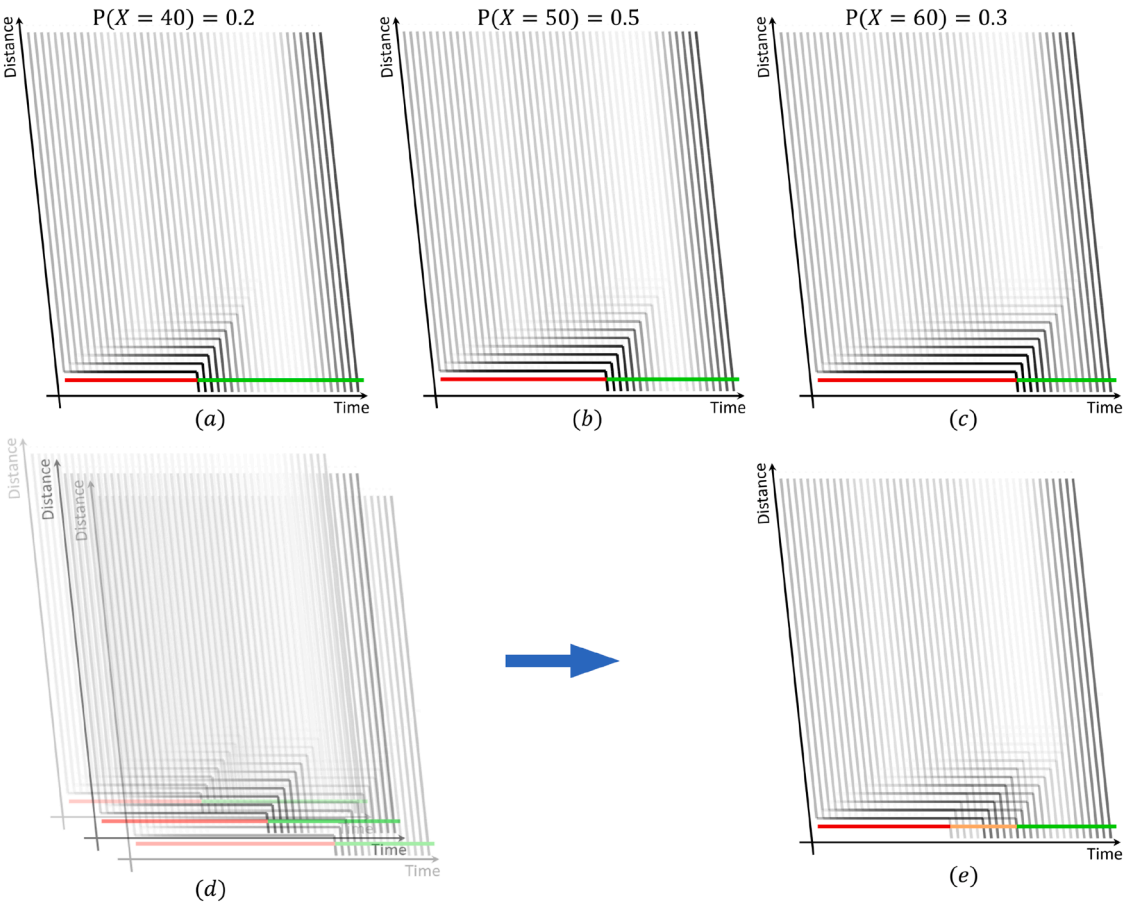


Fig. 7. Visualization of the weighted PTS diagram.

where $p(x)$ is the PMF of actual green start times estimated from Algorithm 1.

Similarly, the average number of stops is

$$n^{\text{est}} = \mathbb{E}_X\left[n(\alpha, \delta^X)\right] = \sum_{x \in \mathcal{G}} p(x) n(\alpha, \delta^x). \tag{20}$$

This weighted PTS model can be visualized by overlaying the corresponding PTS diagrams with $p(X)$ as the weight. In the example illustrated by Fig. 7, this movement has known arrivals and the cycle length is 90 s. There are three possible green start times at 40 s, 50 s, and 60 s. Fig. 7(a)–(c) display the PTS diagrams corresponding to green start times of 40, 50, and 60 s, respectively. Fig. 7(d) shows the weighting of these PTS diagrams according to their respective probabilities, using their probabilities as opacity levels. Fig. 7(e) shows the result of superimposing these diagrams to represent the combined probability of vehicle presence at different times within the cycle.

This approach is valid because the opacity of line segments in the PTS diagram denotes the probability of a vehicle's presence, and the weight $P(X)$ is itself a probability. Consequently, the combined visualization yields a well-defined probability representation. In the overlapping results, the opacity still indicates the likelihood of a vehicle's presence, but now it incorporates the distribution of different green start times. This directly reflects the law of total probability $\mathbb{P}(V) = \sum_{x \in \mathcal{G}} \mathbb{P}(X = x)\mathbb{P}(V | X = x)$, where V is the random variable representing the presence of a vehicle.

4.3. Model calibration

The model calibration process is performed by matching the model-estimated delay in Eq. (19) with the observed delay d^{obs} (Wang et al., 2023) calculated from real-world trajectory data. In Eq. (19), $p(x)$ is estimated by Algorithm 1 and δ^X is also known. The only unknowns are the arrivals α. These arrivals can be obtained by scaling the observed arrival profile according to the penetration rate θ (Wang et al., 2024), so d^{est} becomes a function of θ, denoted $d^{\text{est}}(\theta)$. The calibration process aims to determine the value of θ.

For a movement m, the calibration problem is formulated as:

$$\min_{\theta_m} \left(d_m^{\text{est}}(\theta_m) - d_m^{\text{obs}}\right)^2 + \lambda\left(\theta_m - \theta^{\text{prior}}\right)^2, \tag{21}$$

Table 1
Simulation settings.

Parameter (unit)	Value	
	Uniform Arrival	Coordinated Arrival
Simulation duration (s)	72000	
Cycle length (s)	90	
Number of lanes	1	
Length of link E1 (m)	250	
$g_{\min}$ of J2 (s)	25	
g of J2 (s)	60	
$P(X)$ of J2	Fig. 9(c)	
Green duration of J1 (s)	90	35
Volume (veh/lane/hr)	360	540
Arrival profile	Fig. 9(a)	Fig. 9(b)

where the second term is a regularization term.

For a corridor with movement set $\mathcal{M}$, we minimize:

$$\min_{\Theta} \sum_{m\in\mathcal{M}} n_m^{\text{obs}} \left(d_m^{\text{est}}(\theta_m) - d_m^{\text{obs}}\right)^2 + \lambda \sum_{m\in\mathcal{M}} \left(\theta_m - \bar{\theta}\right)^2, \tag{22}$$

where n_m^{obs} is the number of observed trajectories in movement m, and $\bar{\theta} = \frac{\sum_{m\in\mathcal{M}} n_m^{\text{obs}}\theta_m}{\sum_{m\in\mathcal{M}} n_m^{\text{obs}}}$ is the weighted average penetration rate across movements.

A key consideration when evaluating the corridor model is that d_m^{est} must be computed sequentially according to the movement order in the corridor, since the departure flow of an upstream movement serves as a part of the arrival flow for the downstream movement.

We solve Eq. (22) with a quasi-Newton method (Zhu et al., 1997).

4.4. Cycle length and offset optimization

After calibrating the model, Bayesian optimization (BO) (Nogueira, 2014) is employed to optimize the cycle length and offsets. The decision variables are the offsets $\boldsymbol{\varphi}$ for the intersections along the corridor and the common cycle length C. Given a cycle length C, each offset $\phi \in \boldsymbol{\varphi}$ is constrained to be an integer in $[0, C-1]$, i.e., $\boldsymbol{\varphi} \in \mathcal{F}^N = \{0, 1, \ldots, C-1\}^N$, where N is the number of intersections.

The cost function $\mathscr{L}$ combines the total control delay, $\tau = \sum_{m\in\mathcal{M}} d_m^{\text{est}}$, and the total number of stops, $\nu = \sum_{m\in\mathcal{M}} n_m^{\text{est}}$, and is defined as a function of cycle length C and offsets $\boldsymbol{\varphi}$:

$$\mathscr{L}_{C,\boldsymbol{\varphi}} = \tau + \eta\nu. \tag{23}$$

In this formulation, each stop is treated as equivalent to η s of delay, enabling the optimization process to minimize both delay and stops simultaneously. This weighting reflects the operational impact of stops and encourages solutions that improve traffic flow efficiency. In practice, the parameter η is selected such that τ and $\eta\nu$ are of the same order of magnitude.

Optimizing this cost function is challenging for two reasons. First, evaluating $\mathscr{L}$ requires running the model multiple times for all movements in both directions of the corridor. Second, gradient information is not available, which makes derivative-based optimization methods unsuitable for this problem. However, because the number of decision variables usually equals the number of intersections in the corridor, the problem is low-dimensional. This makes Bayesian optimization (BO) a suitable and tractable approach.

In the Bayesian optimization (BO) framework, a Gaussian Process (GP) is used to construct a posterior distribution that models the cost function. At each iteration, the optimizer selects a set of offsets for evaluation, updates the GP posterior with the resulting cost values, and then chooses the next set of offsets expected to yield the lowest cost, iteratively refining the solution.

To improve computational tractability, we employ a nested optimization strategy that separates the problem into an inner problem and an outer problem. In the inner problem, we optimize offsets for a given cycle length. In the outer problem, we select the best cycle length from a predefined set $\mathcal{C}$ based on the optimal offsets obtained from the inner problem:

$$\textbf{Inner problem:} \quad \boldsymbol{\varphi}^*(C) = \arg\min_{\boldsymbol{\varphi}\in\mathcal{F}^N} \mathscr{L}_{C,\boldsymbol{\varphi}} \tag{24}$$

$$\textbf{Outer problem:} \quad C^* = \arg\min_{C\in\mathcal{C}} \mathscr{L}_{C,\boldsymbol{\varphi}^*(C)}. \tag{25}$$

The size of the predefined cycle length set $\mathcal{C}$ is usually small because cycle lengths are typically multiples of 10 s, with the lower and upper bounds determined by pedestrian minimum green times and link lengths. As a result, this formulation not only reduces the computational burden but also yields implementable optimal solutions. In addition, it provides flexibility, as offset constraints can be easily incorporated, for instance, by fixing the offsets of certain intersections while optimizing others. These advantages make the approach practical for real-world applications.

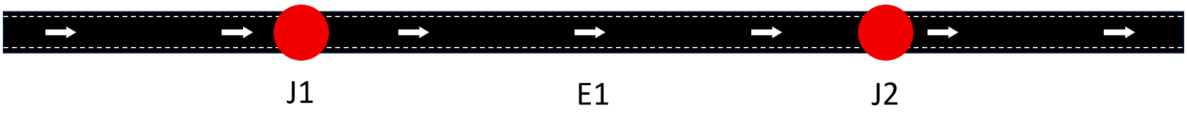


Fig. 8. Road segment in the SUMO simulator. Uniform traffic flow is generated from the far left of the road segment. Intersection J1 is the upstream intersection, and intersection J2 operates with a stochastic green start time to simulate a main direction of coordinated actuated control.

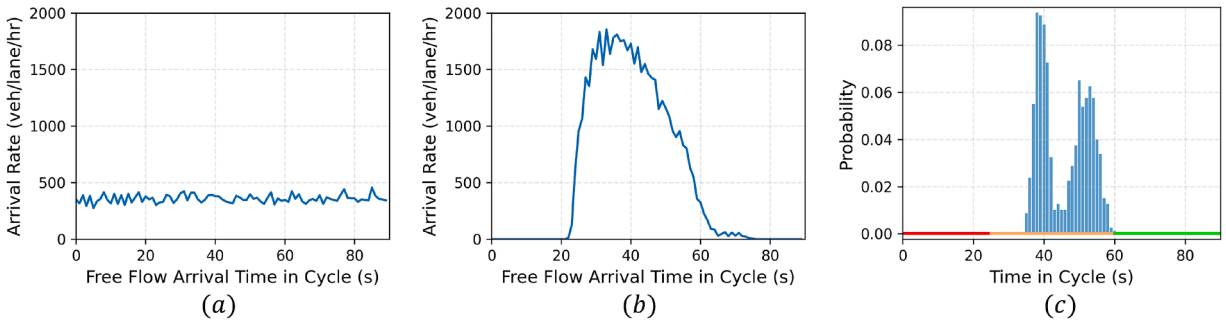


Fig. 9. Generated arrival profiles and green start PMF for J2. (a) Arrival profile under uniform arrivals. (b) Arrival profile under coordinated arrivals. (c) Green start time PMF at J2.

5. Numerical experiments

To comprehensively validate the proposed methodology and quantify its robustness against data scarcity, numerical experiments were conducted using the traffic simulator SUMO (Lopez et al., 2018). This section describes the simulation setup followed by a discussion of the experimental results.

5.1. Simulation setup

We consider two representative arrival profiles commonly observed along corridors: *uniform arrivals*, typical of the upstream boundary of a corridor, and *coordinated arrivals*, where traffic flow is regulated by an upstream traffic signal.

As illustrated in Fig. 8, the simulation network consists of two signalized intersections. For the uniform arrival scenario, the traffic signal at J1 is set to all-green. For the coordinated arrival scenario, J1 operates under fixed-time control. The analysis focuses on the downstream intersection J2 and its connecting link E1. Intersection J2 is configured to operate under coordinated actuated control with stochastic green start times to replicate the ERG phenomenon. Table 1 summarizes the detailed simulation parameters.

To evaluate the impact of data availability on estimation accuracy, we test penetration rates of 5%, 10%, 20%, 50%, 75%, and 100%. For penetration rates below 100%, each experiment is repeated with 100 independent random seeds to capture the uncertainty associated with the stochastic sampling of probe vehicles.

5.2. Results analysis

We first evaluate the accuracy of the estimated green start time distribution. The Jensen–Shannon divergence (JSD) is used to quantify the discrepancy between the estimated PMF and the ground-truth PMF. A JSD value of 0 indicates identical distributions, while a value of 1 indicates no overlap of the two distributions. In practice, a JSD below 0.05 is commonly regarded as indicating very close agreement.

As shown in Fig. 10, for both arrival profiles, the estimated green start PMF closely matches the ground truth even at a penetration rate of 5%. To validate the contribution of non-stopped vehicle data, we conduct an ablation study comparing the use of only Category I observations against the full model using both Category I and Category II observations. Incorporating both categories consistently yields lower JSD values across all penetration rates. This result empirically confirms that the proposed likelihood formulation effectively extracts useful information from censored observations (non-stopped trajectories), thereby enhancing estimation accuracy.

In corridor traffic flow modeling, accurate reconstruction of upstream departures is a prerequisite for predicting downstream arrivals. After estimating the green start distribution, we compare three modeling approaches: (i) the weighted PTS model using the ground truth green start distribution, $\mathbb{E}_X[\mathrm{PTS}(\alpha, \delta^X)]$; (ii) the weighted PTS model using the estimated distribution under 5% penetration; and (iii) a single PTS model constructed using only the mean green start time, $\mathrm{PTS}(\alpha, \delta^{\mathbb{E}[X]})$, a baseline approach similar to Skabardonis (1996).

Fig. 11 illustrates two key findings. First, the proposed weighted PTS model matches the ground-truth departure profile, even when the green start PMF is estimated from low-penetration data. Second, relying solely on the mean green start time leads to substantial

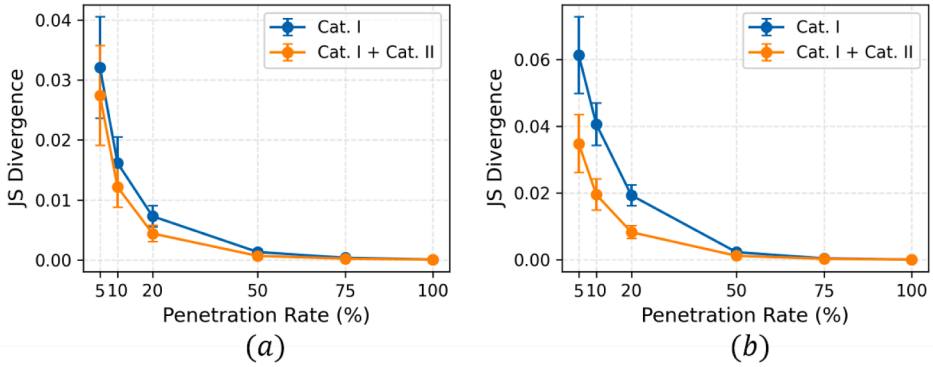


Fig. 10. Jensen–Shannon divergence (JSD) between the ground-truth green start PMF and the estimated PMF under different penetration rates. (a) JSD comparison under uniform arrivals. (b) JSD comparison under coordinated arrivals.

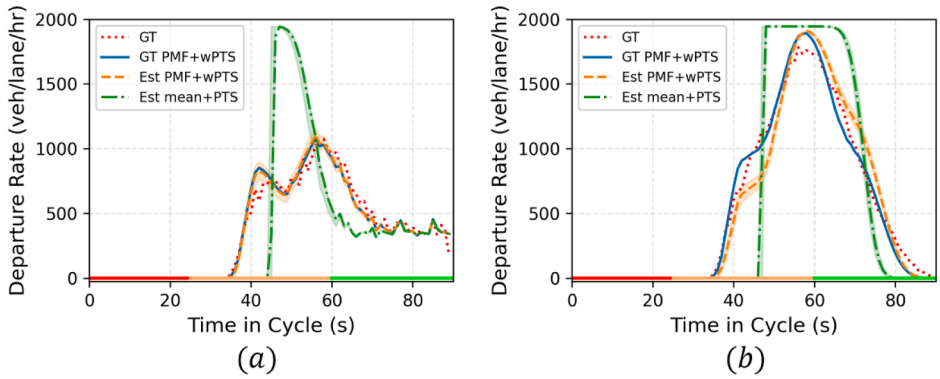


Fig. 11. Departure profiles at J2. The red dotted line shows the ground truth. The solid blue line corresponds to the weighted PTS model using the ground-truth green start PMF. The dashed orange line shows the weighted PTS model using the estimated green start PMF (5% penetration). The dash-dotted green line shows the PTS model constructed using only the mean green start time. The shaded region around the orange and green curves denotes the 10th–90th percentile interval computed over 100 random seeds.

errors; specifically, the baseline method produces a "step-like" fixed-time departure curve in the departure profile and fails to capture the smoothing effect caused by the stochastic signal operations. This comparison highlights the necessity of incorporating the full distribution of green start times to ensure reliable corridor-level traffic predictions.

Overall, these numerical experiments demonstrate that (i) the proposed method is robust to low penetration rates, and (ii) explicitly modeling the green start distribution is essential for capturing the dynamics of coordinated actuated control.

6. Case study

The study was conducted at seven intersections along a 2.5-mile segment of Telegraph Rd, from County Center Dr. to Voorheis Rd. in Pontiac, MI, as shown in Fig. 12(a). The corridor operates under coordinated actuated control. In the northern half of the segment, from County Center Dr to Elizabeth Lake Rd, both the northbound and southbound directions have two lanes; beyond this point, each direction expands to three lanes. Fig. 12(b) shows the spacing between intersections (in meters). Intersections with street names in red indicate locations where at least one non-coordinated movement operates under minimum or no recall, making an ERG scenario possible for the coordinated movements.

In the following sub-sections, we first use a single movement in this corridor to demonstrate how we estimate the empirical green start PMF and construct the corresponding weighted probabilistic time-space diagram. We then present the overall workflow of our approach to calibrate and optimize the entire corridor. Both analyses are based on real-world data, and the optimized signal plan was implemented in the field.

6.1. Demonstration of ERG modeling on a single movement

In this section, we will provide a comprehensive analysis of one movement in the study corridor: the southbound through movement at Telegraph Road and Vanguard Dr., shown in Fig. 13(a). Fig. 13(b) illustrates the phase structure of this intersection. According to the ring-and-barrier diagram, phases 2 and 6 are the coordinated phases, with our target movement operating under phase 6. Phases 4

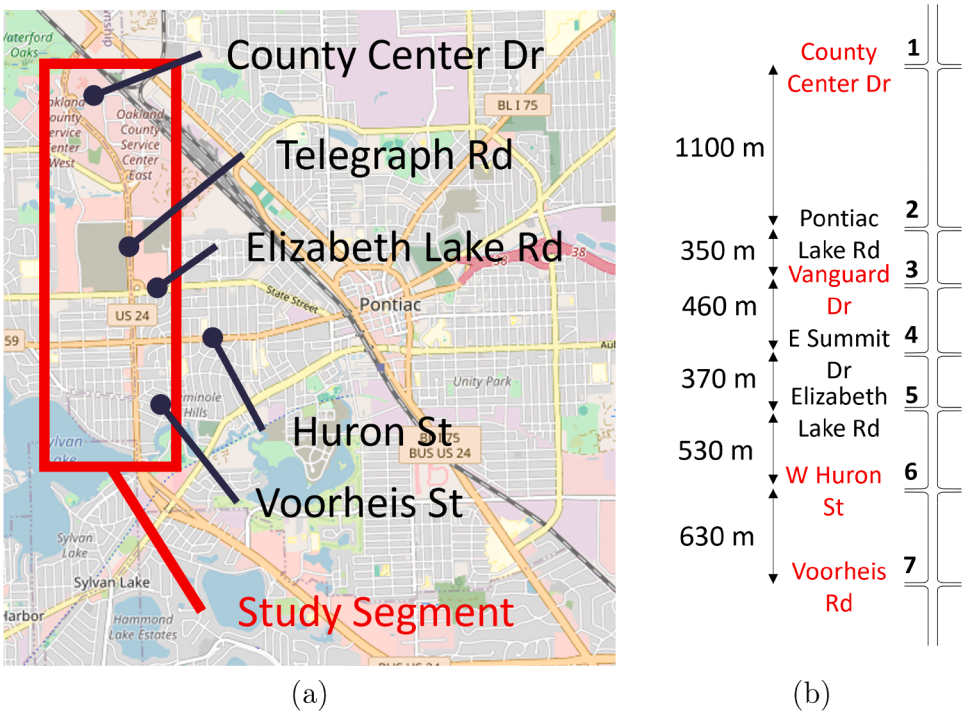


Fig. 12. Study area: a 2.5-mile segment of Telegraph Road in Pontiac, MI. (a) Study segment on map. (b) Seven intersections along the corridor. Intersections shown in red indicate that at least one non-coordinated movement operates under minimum or no recall.

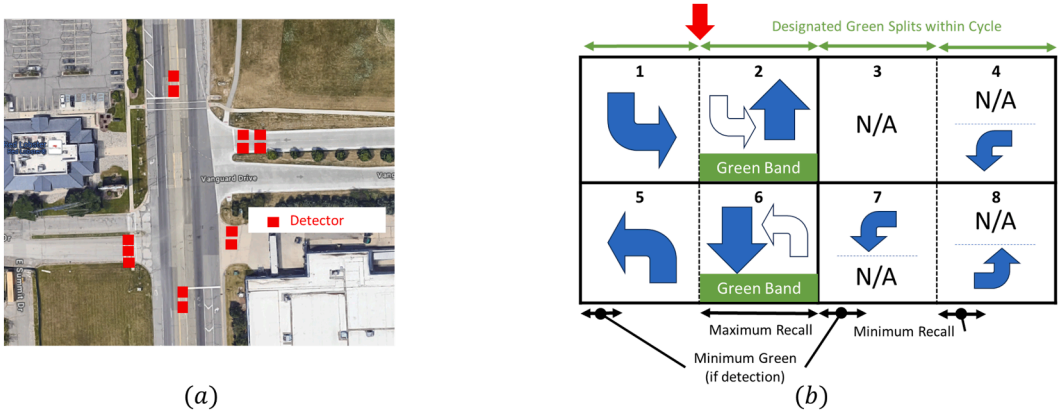


Fig. 13. Intersection of Telegraph Road and Vanguard Drive. (a) Satellite photo of intersection with detectors' position. (b) Phase structure of this intersection. The red arrow means the coordination start point.

and 8 are set at minimum recall while phases 1, 5, and 7 are operated under no recall. Any extra green time from phase skips or gap-outs is allocated to the coordinated phases (2 and 6), causing their actual green start times to occur earlier than the designated coordination start point. For the target movement, the corresponding signal split is summarized in Table 2.

We obtain the green start time PMF from trajectory data using the method described in Section 4.1. The trajectory dataset spans from October 2, 2023, to October 20, 2023 (three weeks in total), and includes only weekdays. The results are shown in Fig. 14. The first column presents the aggregated time-space diagram based on real-world trajectory data. The second column displays the estimated PMF of the green start times, and the third column shows the weighted probabilistic time-space diagram. Each row corresponds to a different time-of-day (ToD) period.

Across all figures, the x-axis represents the time within the cycle, and the signal plan is indicated on the color bar at $y = 0$: red denotes the red phase, orange the potential ERG range, and green the designated green split, followed by yellow and red for clearance. For the time-space and weighted probabilistic time-space diagrams, the y-axis represents the distance to the intersection.

Table 2
SPaT of the southbound through movement.

ToD	Time Range	Cycle	g_{min}	g
Morning Peak	07:00-10:00	120	16.1	66
Mid-Day	10:00-15:00	100	16.1	60
Evening Peak	15:00-19:00	120	16.1	69

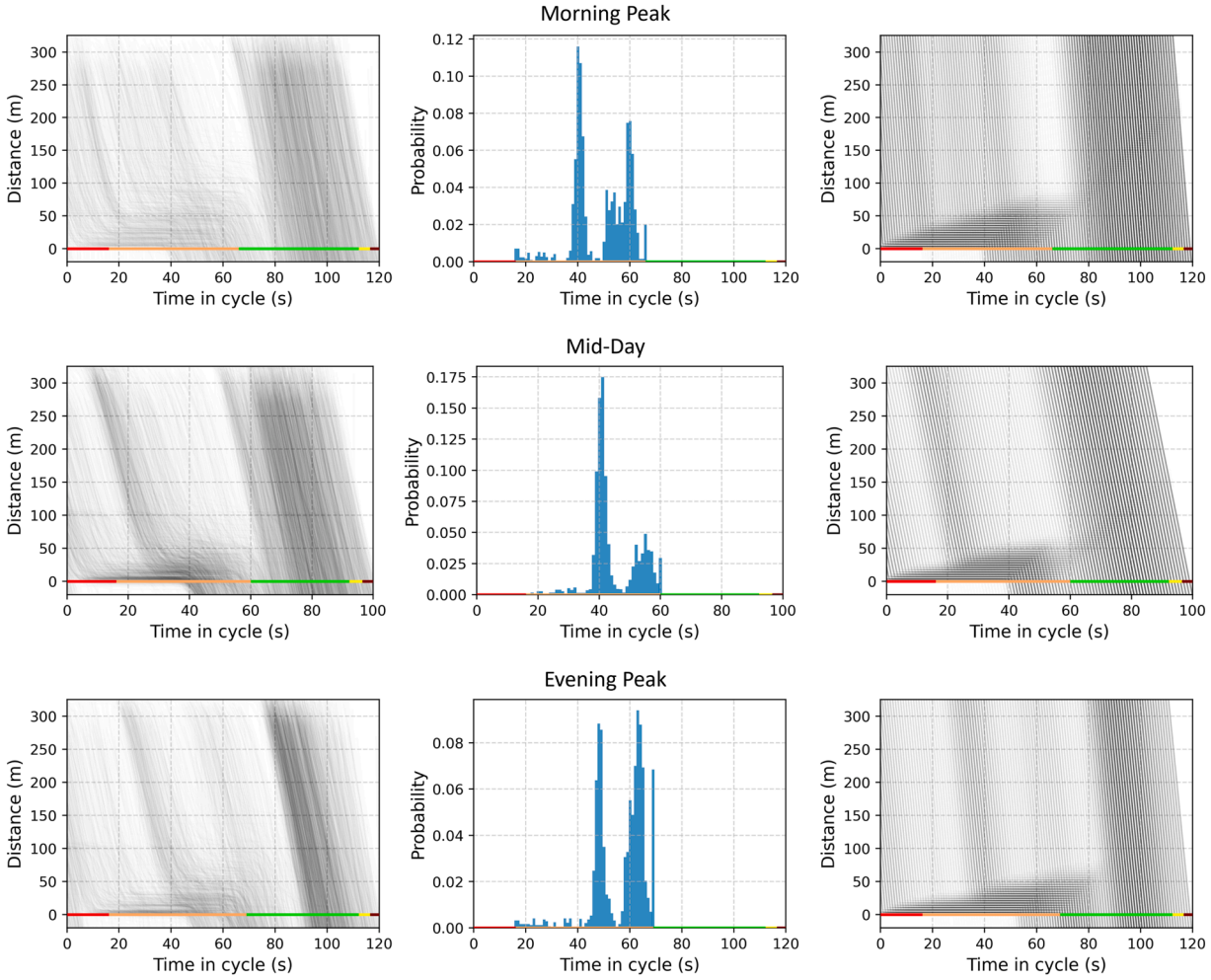


Fig. 14. Modeling the movement with ERG in three ToDs. The orange segment in the signal bar represents the possible range of green start time.

The first and third columns are visually similar. In particular, when focusing on the vehicle departures area where $y < 0$, the departure patterns match closely. This comparison indicates that our method effectively captures departures within the ERG range, enabling the weighted PTS model to represent the traffic state accurately. Because controller event logs or equivalent field observations of the true green start times are unavailable at this site, a direct field comparison of the estimated PMF is not possible. Therefore, the estimation method itself is directly validated in Section 5, while the agreement between the first and third columns here provides indirect support for the estimated PMF in the real-world case study.

Furthermore, examining only the second column of Fig. 14 reveals subtle variations in green start patterns across different ToD periods. The two peaks are the result of some of the non-coordinated phases with no recall being serviced. Table 3 provides a quantitative assessment of ERG patterns for these periods. The morning peak exhibits the highest occurrence rate, with ERG not occurring in only 2% of cycles. All ToD periods show a normalized average ERG duration of 23% to 35%, with a mild degree of variation. From these metrics, we can conclude that the detection systems are operational, and the configuration of the controller is appropriate.

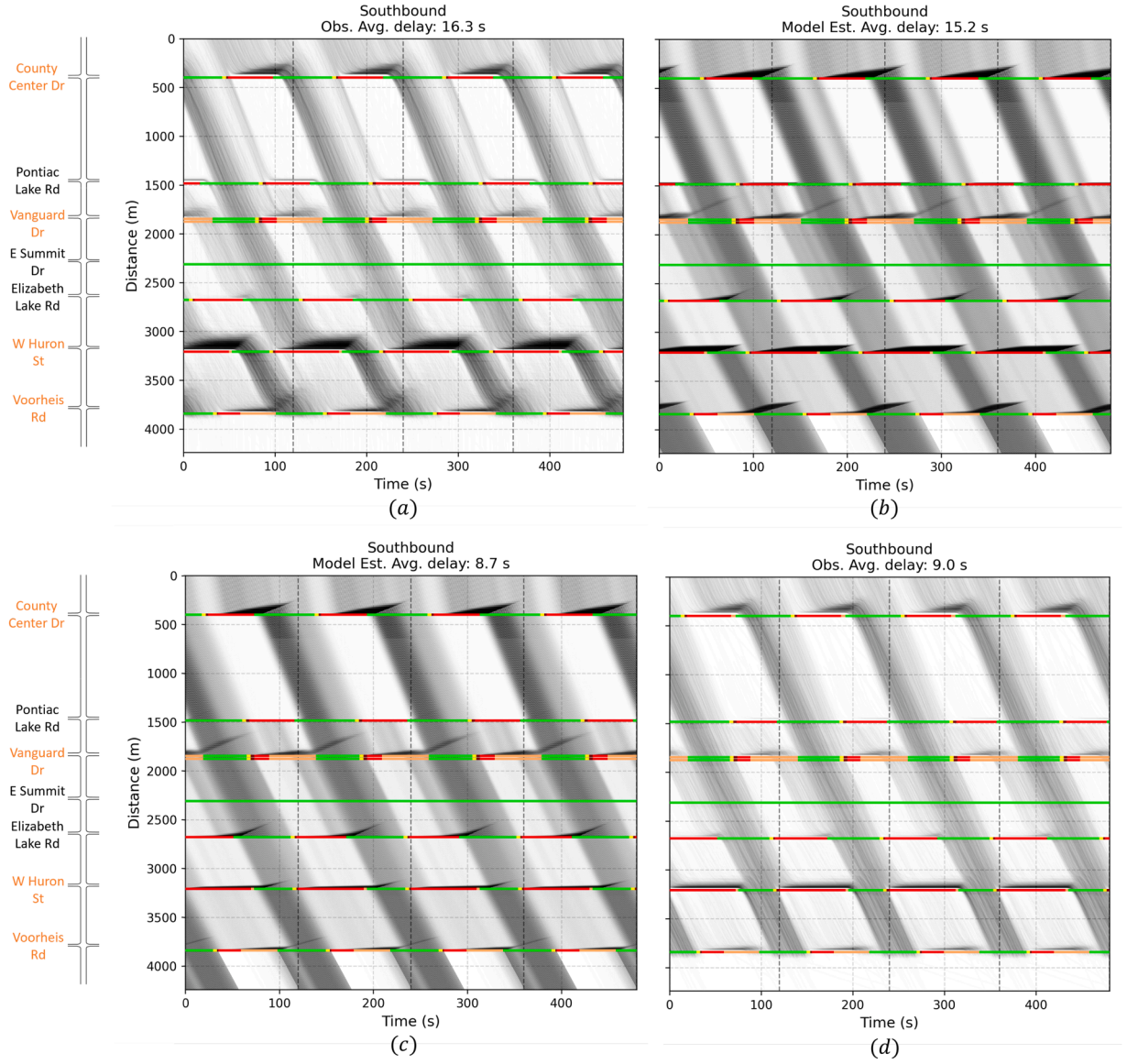


Fig. 15. Time-space (TS) diagram and weighted probabilistic time-space (PTS) diagram for the corridor with ERG intersections during the southbound morning peak (07:00–09:00). Intersection names shown in orange indicate those operating under coordinated actuated control. (a) Original TS diagram from real-world trajectory data (Oct. 2–20, 2023). (b) Calibrated weighted PTS model using the same data as (a). (c) Predicted weighted PTS diagram from the calibrated model under the optimized signal plan. (d) TS diagram from real-world trajectory data after deploying the optimized signal plan on Mar. 4, 2025, using data from Mar. 10–28, 2025.

6.2. *Corridor-scale implementation and performance evaluation*

As outlined in Fig. 3, our workflow consists of four main steps. In the previous section, we presented the results of the first two steps for a single movement. In this section, we extend the analysis to the entire corridor and present the results of the latter two steps.

After constructing the weighted PTS model for each movement in the corridor, as described earlier, we calibrate them jointly as a corridor-level model using real-world trajectory data, following Eq. (22) with $\lambda = 10^3$. The calibrated model is then used for signal optimization based on Eqs. (24)-(25), with $\eta = 10$ and $C = \{80, 90, 100, 110, 120\}$, yielding an optimized signal plan. The calibration process used trajectory data collected from October 2 to October 27, 2023 (weekdays only), and the optimization was performed with respect to the signal plan operating during the same period. This optimization process not only generates the plan but, more importantly, provides a prediction of the traffic conditions expected after deployment. The optimized plan was subsequently implemented along the corridor.

Table 3
ERG-related metrics for this movement.

ToD	R_{occu}	R_{avg}	R_{std}
AM	98.0%	34.7%	21.6%
MD	97.1%	34.8%	17.7%
PM	93.2%	23.7%	18.9%

Table 4
Estimated average coordinated movement volume across evaluation periods under the original, consultant, and proposed signal plans.

ToD	Direction	Avg. volume (veh/h)		
		Original	Consultant	Proposed
		Oct. 2023	Mar. 2024	Mar. 2025
AM	Northbound	947	858	979
	Southbound	1520	1535	1592
	Both	1234	1197 (−3.0%)	1286 (+4.2%)
MD	Northbound	1286	928	1176
	Southbound	1097	1071	1084
	Both	1192	1000 (−16.1%)	1130 (−5.2%)
PM	Northbound	1739	1508	1635
	Southbound	970	957	975
	Both	1355	1233 (−9.0%)	1305 (−3.7%)

Note: Percent changes in parentheses are relative to the corresponding value in the period running the original plan (Oct. 2023).

We first qualitatively evaluate the calibration results and the improvement achieved by the optimization. Fig. 15 shows the time-space diagram and weighted probabilistic time-space diagram for the southbound direction during the morning peak hour (07:00–09:00). The relationship between these four subfigures illustrates successive steps in the process:

- The comparison between (a) and (b) confirms that the calibrated PTS model accurately reproduces the real-world time-space diagram.
- The changes between (b) and (c) demonstrate that the model responds appropriately to modifications in the signal plan.
- The agreement between (c) and (d) indicates that the model can predict the post-implementation traffic state with high accuracy.
- Finally, the improvement from (a) to (d) shows that, after optimization, traffic flow becomes noticeably smoother, with the observed average delay reduced by more than 44%.

In addition to the original field plan and our optimized plan, we include an additional practical benchmark: a consultant plan optimized for the same corridor by an independent consulting firm. The firm used Synchro/SimTraffic to develop the optimized plan to reduce stops and delays. For clarity, in the following discussion, the *original plan* refers to the field signal timing plan operating during this October 2023 study period, which also serves as the baseline plan for our optimization. The *consultant plan* refers to the signal timing plan optimized by the consulting firm; it was deployed on February 21, 2024, and evaluated using probe vehicle data from March 4 to March 28, 2024. The *proposed plan* refers to the plan optimized using our proposed framework; it was deployed on March 4, 2025, and evaluated using data collected from March 10 to March 28, 2025.

Since the three signal plans were evaluated in different periods, we first examine whether the underlying demand levels are comparable before discussing the performance results. The overall arrivals can be recovered based on Θ^* obtained from Eq. (22) after the model calibration process. By summing these arrivals, we estimate the average coordinated movement volumes under the original, consultant, and proposed signal plans. The comparison is presented in Table 4. Compared with the original-plan period, the proposed-plan period shows relatively small changes in coordinated movement volumes, whereas the consultant-plan period exhibits larger reductions, especially during MD and PM. Therefore, the comparison with the consultant plan should be interpreted as an additional practical benchmark rather than a controlled same-demand experiment. Even so, Table 4 provides useful context for interpreting the performance differences reported below.

Table 5 compares corridor travel time, average movement control delay, and average movement number of stops under the original, consultant, and proposed signal plans. These performance metrics are calculated from probe vehicle data using the method described in Wang et al. (2023). Relative to the original plan, both retimed plans improve corridor operations. The proposed plan shows slightly lower all-day values for all three metrics and performs best during the AM period across all three metrics. The consultant plan performs better for PM control delay and MD number of stops. Overall, the proposed plan reduces corridor travel time by 9.9%, average control delay by 17.2%, and average number of stops by 20.7% relative to the original plan. Taken together, this additional comparison suggests that the proposed framework can achieve a performance level comparable to, and slightly better than, the consultant plan on the all-day aggregate metrics. Beyond the performance comparison, the proposed framework is also more scalable in practice because it does not require direct access to controller logs, which were not available for this corridor; using probe vehicle data, the

Table 5
Performance evaluation of the original, consultant, and proposed signal plans.

Metrics	Plan	All day	AM	MD	PM
Avg. Travel Time (s)	Original	277	288	257	309
	Consultant	256 (−7.6%)	256 (−11.3%)	256 (−0.2%)	264 (−14.3%)
	Proposed	**250 (−9.9%)**	**241 (−16.3%)**	**252 (−1.8%)**	**263 (−14.8%)**
Avg. Control Delay (s)	Original	13.5	14.9	11.3	16.8
	Consultant	11.5 (−15.0%)	11.4 (−23.1%)	12.1 (+6.6%)	**12.5 (−25.5%)**
	Proposed	**11.2 (−17.2%)**	**10.8 (−27.3%)**	**11.0 (−2.7%)**	12.9 (−23.2%)
Avg. # of Stops	Original	0.28	0.29	0.26	0.32
	Consultant	0.23 (−18.3%)	0.21 (−28.4%)	**0.23 (−11.6%)**	0.26 (−18.7%)
	Proposed	**0.22 (−20.7%)**	**0.19 (−34.5%)**	0.25 (−5.5%)	**0.24 (−25.8%)**

Note: Percent changes in parentheses are relative to the original plan. For each metric and time period, the best value is highlighted in bold.

Table 6
ERG-related metrics for the coordinated movements under coordinated actuated control along the study corridor.

Intersection	Direction	ToD	E_{max} (s)	R_{occu}	R_{avg}	R_{std}
County Center Dr	Southbound	AM	4.9	0.6%	0.2%	3.6%
		MD	9.9	1.2%	0.5%	5.5%
		PM	11.9	1.7%	1.1%	9.1%
	Northbound	AM	4.9	0.7%	0.3%	4.9%
		MD	9.9	5.3%	2.8%	12.4%
		PM	11.9	2.0%	1.5%	11.0%
Vanguard Dr	Southbound	AM	49.9	98.0%	34.7%	21.6%
		MD	43.9	97.1%	34.8%	17.7%
		PM	52.9	93.2%	23.7%	18.9%
	Northbound	AM	49.9	98.5%	40.3%	25.9%
		MD	43.9	97.1%	36.1%	19.4%
		PM	52.9	91.4%	30.4%	24.2%
W Huron St	Southbound	AM	3.2	0.6%	0.5%	5.9%
		MD	8.2	6.7%	2.1%	9.8%
		PM	18.2	4.9%	1.8%	10.1%
	Northbound	AM	3.2	0.6%	0.4%	5.1%
		MD	8.2	5.9%	1.9%	10.0%
		PM	18.2	15.5%	4.2%	13.8%
Voorheis Rd	Southbound	AM	39.0	50.9%	10.9%	17.2%
		MD	37.0	73.7%	29.3%	27.5%
		PM	54.0	64.6%	22.0%	23.8%
	Northbound	AM	31.7	45.3%	15.9%	26.4%
		MD	21.7	68.7%	25.2%	22.1%
		PM	22.7	44.3%	14.7%	22.3%

seven-intersection corridor across three ToD periods can be calibrated and optimized within one day, whereas traditional retiming using Synchro/SimTraffic typically requires a longer process of field data collection, model development, calibration, and optimization that can take months in practice.

Table 6 presents additional ERG-related metrics for the coordinated movements of all four intersections operating under coordinated actuated control along the study corridor. Except for County Center Dr., the other three intersections include leading actuated left-turn phases preceding the coordinated phases in both the southbound and northbound directions. Consequently, the ERG metrics for the two directions may differ.

The results reveal distinct traffic patterns across intersections. For instance, at Vanguard Dr., ERG occurs in nearly every cycle, with consistently high ERG duration usage across all time-of-day (ToD) periods, indicating persistently low non-coordinated traffic volumes. In contrast, Voorheis Rd. exhibits a more dynamic pattern: ERG is less frequent during peak periods and more prevalent during off-peak periods, suggesting that the non-coordinated phases follow a peak-off-peak demand cycle.

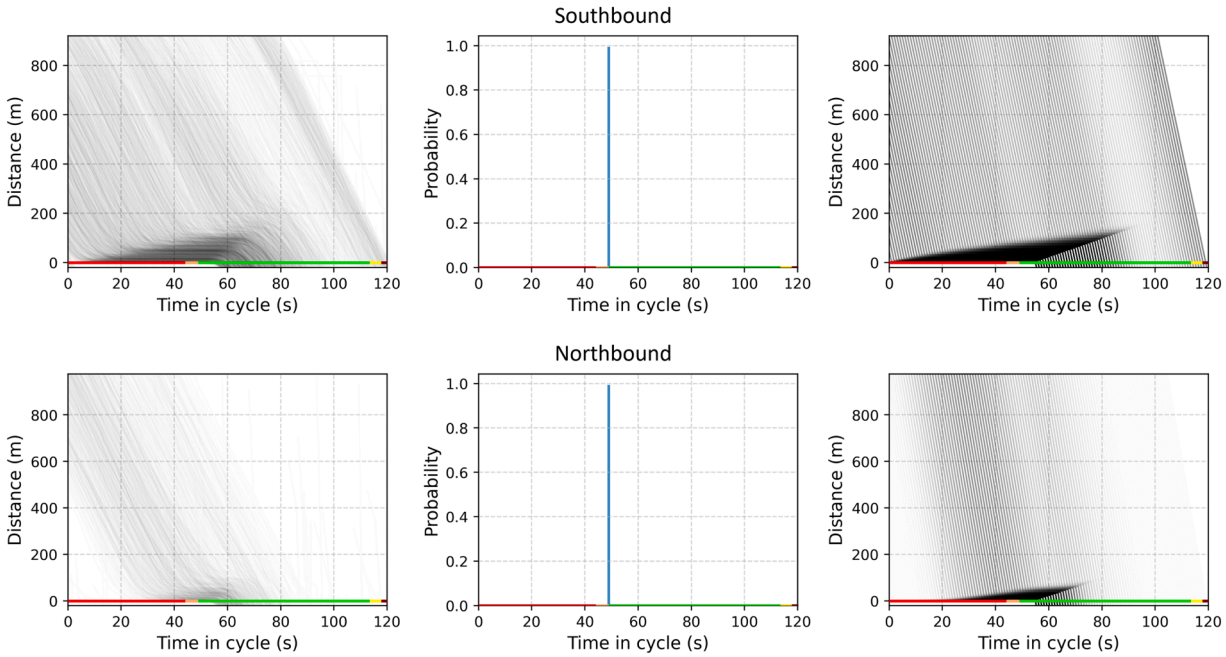


Fig. 16. Southbound and northbound movements at County Center Dr during the AM period. The first column shows time-space diagrams, the second column shows the PMFs of actual green start times, and the third column shows the corresponding weighted probabilistic time-space diagrams.

Beyond revealing traffic patterns, these metrics can also indicate potential loop-detector malfunctions. For example, both R_{occu} and R_{avg} for the southbound and northbound approaches at County Center Dr. are extremely low across all ToD periods, suggesting possible detector or controller failures. As shown in Fig. 16, almost no trajectories pass the intersection before the designated green start, corroborating this diagnosis. We reported this observation to the agency, which subsequently confirmed that the vehicle detectors at this intersection were indeed broken. These findings highlight the potential of trajectory data to serve as an effective, low-cost infrastructure monitoring tool.

7. Conclusion

In this paper, we proposed a framework to improve coordinated actuated control performance by incorporating the effect of "Early Return to Green" (ERG). The framework consists of four main steps: (1) estimating the distribution of actual green start times from trajectory data using an EM algorithm; (2) using the estimated distributions as weights to construct a weighted probabilistic time-space (PTS) model; (3) calibrating this model with real-world trajectory data; and (4) performing cycle length and offset optimization based on the calibrated model. The approach is concise, straightforward, and ready for large-scale implementation.

The framework makes two primary methodological contributions. First, we formulate a nonparametric maximum likelihood estimator for the distribution of actual green start times based on trajectory data and develop an EM algorithm to solve it. Second, we extend the original PTS diagram, which is designed for intersections with fixed-time control, to a weighted PTS model that accommodates vehicle-actuated control. The weighted PTS model effectively captures the traffic state of a movement influenced by ERG.

Another contribution of this work is the introduction of three ERG-related metrics that quantify the impact of ERG. These metrics are intuitive, computationally simple, and applicable beyond trajectory data; for example, they can also be derived from controller logs. Since vehicles are the end users of traffic signals, trajectory-based metrics provide a comprehensive assessment that evaluates both control logic and detector hardware. These metrics reveal side-street traffic patterns and can serve as indicators for detecting detector malfunctions or for fine-tuning vehicle-actuated control settings in a timely and cost-effective manner.

Numerical experiments utilizing SUMO confirm that the proposed framework accurately estimates green start distributions even at low penetration rates and significantly outperforms baseline models relying on mean green start times. The framework was validated in a real-world case study on a seven-intersection corridor, four of which have vehicle-actuated movements on the side-street phases. We first illustrated the methodology on a single movement by estimating the empirical green start PMF and constructing its weighted PTS model. At the corridor level, we applied the framework to optimize cycle length and offsets under vehicle-actuated control. A before-and-after analysis of the real-world implementation confirmed the effectiveness of the method. Moreover, the calibration and optimization process can be fully automated, allowing optimization of a seven-intersection corridor across three time-of-day (ToD) periods within a single day. Finally, we demonstrated the application of ERG-related metrics in practice. Together, these capabilities highlight the potential of the proposed framework to outperform traditional optimization methods.

CRediT authorship contribution statement

Zihao Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Xingmin Wang:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Zachary Jerome:** Writing – review & editing, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Henry X. Liu:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to polish the writing of this paper. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

None.

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Data availability

The authors do not have permission to share data.

Appendix A. Probabilistic Time-Space (PTS) Model

The probabilistic time-space (PTS) model (Wang et al., 2024) describes the stochastic queueing process at signalized intersections based on a simplified version of Newell's car-following model (Newell, 2002). Each vehicle alternates between two states: stopped or free-flow state at speed v_f. The traffic dynamics are represented in discrete time with interval Δt.

The model is formulated in Newellian coordinates, where the horizontal axis denotes free-flow arrival time and the vertical axis represents the cumulative number of vehicles. In this space, the stochastic queue evolution follows the discrete-time dynamic:

$$X(t) = X(t-1) + A(t) - B(t), \tag{A.1}$$

where $X(t)$ is the number of stopped vehicles after time t, $A(t)$ is the binary vehicle arrival (1 if a vehicle arrives), and $B(t)$ is the binary vehicle departure (1 if a vehicle departs). The departure occurs only when both the queue is non-empty and the signal is green:

$$\mathbb{P}(B(t) = 1) = \mathbb{P}(X(t) \geq 1, S(t) = 1), \tag{A.2}$$

where $S(t) \in \{0, 1\}$ represents the signal state (0 = red, 1 = green).

Let $x(t, k)$ denote the probability that the queue length equals k at time t, and $\boldsymbol{x}(t, :)$ be the queue-length PMF. The transition of this PMF is given by:

$$\begin{aligned} \boldsymbol{x}(t, :) &= f(\boldsymbol{x}(t-1, :), a(t), S(t)) \\ &= \begin{cases} x'(t, k+1) = x(t-1, k) \cdot a(t) + x(t-1, k+1) \cdot (1 - a(t)) \\ x(t, k) = x'(t, k+1) \cdot S(t) + x'(t, k) \cdot (1 - S(t)), \ \forall k \geq 1 \\ x(t, 0) = x'(t, 1) \cdot S(t) + x'(t, 0) \end{cases} \end{aligned} \tag{A.3}$$

where $a(t) = \mathbb{P}(A(t) = 1)$ is the Bernoulli arrival rate and $f(\cdot)$ denotes the transition mapping function.

Finally, the PTS diagram projects the stochastic point-queue process defined by Eqs. (A.1)–(A.3) back to the time-space domain, producing a probabilistic representation of vehicle trajectories. This model establishes a bidirectional mapping between the microscopic trajectory space and macroscopic queue dynamics, enabling efficient computation of traffic performance metrics.

For brevity, the over-saturated scenario (with residual queues) is omitted here; its treatment follows directly from the extended PTS formulation in the supplementary material of Wang et al. (2024).

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